

Geodesics on a Cone

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The Geodesic Equation

Let $U \subseteq \mathbb{C}$ be an open region. In the complex coordinate z , a conformal metric has the form $g = u(z)|dz|^2$ where u is called a *weight function*. Here $|dz|^2$ represents the standard complex dot product. At a point $p \in U$ and a vector w starting at p , the norm of w with respect to g is $|w|_g = \sqrt{u(p)}|w|$, where $|w|$ is the standard norm. Note that we've switched from the earlier convention $g = u^2|dz|^2$ to match the M.R.S. paper.

The connection induced by g is given by $\nabla = d + \partial \log u$. Evaluating this connection on a function f gives

$$\nabla f = df + (\partial_z \log u) f dz \quad (1)$$

In the case where $\gamma : [0, T) \rightarrow U$ is a geodesic, we can extend the derivative $\dot{\gamma}$ to a vector field on a tubular neighborhood of $\text{im } \gamma$. Since the tangent bundle of U is trivial, a vector field is the same as a complex function. Thus

$$0 = \nabla_{\dot{\gamma}} \dot{\gamma} = \ddot{\gamma} + (\partial_z \log u) \dot{\gamma} \cdot \langle dz, \dot{\gamma} \rangle = \ddot{\gamma} + (\partial_z \log u) \dot{\gamma}^2, \quad (2)$$

where $\langle dz, \dot{\gamma} \rangle$ is the duality pairing. Our weight function will be given in terms of polar coordinates (r, θ) . Recall

$$\partial_z = \frac{e^{-i\theta}}{2r} (r \partial_r - i \partial_\theta)$$

so (2) becomes

$$0 = \nabla_{\dot{\gamma}} \dot{\gamma} = \ddot{\gamma} + \frac{e^{-i\theta}}{2} (\partial_r \log u - i r \partial_\theta \log u) \dot{\gamma}^2. \quad (3)$$

We still consider $\dot{\gamma}$ and $\ddot{\gamma}$ to be complex numbers.

Approximating Geodesics on a Cone

Let p and v be the initial position and velocity for the geodesic, both given as complex numbers. Denoting $\nabla_z = \frac{e^{-i\theta}}{2} (\partial_r \log u - i r \partial_\theta \log u)$, we will apply Euler's method to

$$\partial_t \dot{\gamma} = -\nabla_z \dot{\gamma}^2. \quad (4)$$

The only difference from our geodesics on the cone is that now we must approximate ∇_z using the finite element data we simulated. To do this, we need to:

- Find the point nearest to $\gamma(t)$, given in polar coordinates;
- Use first differences to approximate ∇_z ;
- Update $\dot{\gamma}$ according to (4);

- Update γ by adding $(\Delta t)\ddot{\gamma}$.